ZX Spectrum 48K System Variables

23552–23733 ($5C00–$5CB5). 2-byte values are stored low byte first: read with `PEEK n+256*PEEK (n+1)`; write with `POKE n,v-256*INT (v/256): POKE n+1,INT (v/256)`. B = size in bytes. X/N column (from the manual): **X** = don't poke, the system might crash; **N** = poking has no lasting effect.

Name	Dec	Hex	B	X/N	Contents
KSTATE	23552	$5C00	8	X	Used in reading the keyboard (two 4-byte key sets)
LAST K	23560	$5C08	1	N	Newly pressed key
REPDEL	23561	$5C09	1		Frames a key must be held before it repeats (starts at 35)
REPPER	23562	$5C0A	1		Frames between repeats of a held key (starts at 5)
DEFADD	23563	$5C0B	2	N	Address of arguments of user-defined function being evaluated, else 0
K DATA	23565	$5C0D	1	N	2nd byte of colour controls entered from keyboard
TVDATA	23566	$5C0E	2	N	Colour, AT and TAB control bytes going to the TV
STRMS	23568	$5C10	38	X	Addresses of channels attached to streams
CHARS	23606	$5C36	2		256 less than address of character set (point to RAM for your own font)
RASP	23608	$5C38	1		Length of warning buzz
PIP	23609	$5C39	1		Length of keyboard click
ERR NR	23610	$5C3A	1		1 less than report code; starts at 255 (IY points here)
FLAGS	23611	$5C3B	1	X	Various flags controlling the BASIC system
TV FLAG	23612	$5C3C	1	X	Flags associated with the television
ERR SP	23613	$5C3D	2	X	Address of machine-stack item used as error return
LIST SP	23615	$5C3F	2	N	Address of return address from automatic listing
MODE	23617	$5C41	1	N	Specifies K, L, C, E or G cursor
NEWPPC	23618	$5C42	2		Line to be jumped to
NSPPC	23620	$5C44	1		Statement in line to jump to (poke NEWPPC then NSPPC to force a jump)
PPC	23621	$5C45	2		Line number of statement being executed
SUBPPC	23623	$5C47	1		Number within line of statement being executed
BORDCR	23624	$5C48	1		Border colour ×8; also lower-screen attributes
E PPC	23625	$5C49	2		Number of current line (with program cursor)
VARS	23627	$5C4B	2	X	Address of variables
DEST	23629	$5C4D	2	N	Address of variable in assignment
CHANS	23631	$5C4F	2	X	Address of channel data
CURCHL	23633	$5C51	2	X	Address of info currently used for input/output
PROG	23635	$5C53	2	X	Address of BASIC program
NXTLIN	23637	$5C55	2	X	Address of next line in program
DATADD	23639	$5C57	2	X	Address of terminator of last DATA item
E LINE	23641	$5C59	2	X	Address of command being typed in
K CUR	23643	$5C5B	2		Address of cursor
CH ADD	23645	$5C5D	2	X	Address of next character to be interpreted
X PTR	23647	$5C5F	2		Address of character after the ? marker

Name	Dec	Hex	B	X/N	Contents
WORKSP	23649	$5C61	2	X	Address of temporary work space
STKBOT	23651	$5C63	2	X	Address of bottom of calculator stack
STKEND	23653	$5C65	2	X	Address of start of spare space
BREG	23655	$5C67	1	N	Calculator's B register
MEM	23656	$5C68	2	N	Address of calculator's memory area (usually MEMBOT)
FLAGS2	23658	$5C6A	1		More flags (bit 3 = CAPS LOCK on)
DF SZ	23659	$5C6B	1	X	Lines in lower part of screen (incl. 1 blank line)
S TOP	23660	$5C6C	2		Number of top program line in automatic listings
OLDPPC	23662	$5C6E	2		Line number to which CONTINUE jumps
OSPCC	23664	$5C70	1		Statement within line to which CONTINUE jumps
FLAGX	23665	$5C71	1	N	Various flags
STRLEN	23666	$5C72	2	N	Length of string-type destination in assignment
T ADDR	23668	$5C74	2	N	Address of next item in syntax/parameter table
SEED	23670	$5C76	2		Seed for RND (set by RANDOMIZE)
FRAMES	23672	$5C78	3		Frame counter, LSB first; +1 every 20 ms
UDG	23675	$5C7B	2		Address of first user-defined graphic
COORDS	23677	$5C7D	2		x (23677) and y (23678) of last point plotted
P POSN	23679	$5C7F	1		33 − column number of printer position
PR CC	23680	$5C80	2	X	Address of next LPRINT position in printer buffer
ECHO E	23682	$5C82	2		33 − column and 24 − line of end of input buffer
DF CC	23684	$5C84	2		Display-file address of PRINT position
DFCCL	23686	$5C86	2		Like DF CC for lower part of screen
S POSN	23688	$5C88	2	X	33 − column (23688) and 24 − line (23689) for PRINT
SPOSNL	23690	$5C8A	2	X	Like S POSN for lower part of screen
SCR CT	23692	$5C8C	1		Scroll counter: 1 more than scrolls before 'scroll?'
ATTR P	23693	$5C8D	1		Permanent current colours (colour statements)
MASK P	23694	$5C8E	1		Transparent colours: 1-bits taken from screen, not ATTR P
ATTR T	23695	$5C8F	1	N	Temporary current colours (colour items)
MASK T	23696	$5C90	1	N	Like MASK P, but temporary
P FLAG	23697	$5C91	1		More flags
MEMBOT	23698	$5C92	30	N	Calculator's memory area
NMIADD	23728	$5CB0	2		Non-maskable interrupt address (read by ROM's NMI code)
RAMTOP	23730	$5CB2	2		Address of last byte of BASIC system area
P RAMT	23732	$5CB4	2		Address of last byte of physical RAM

Handy POKEs and notes

- `POKE 23609,n` (PIP) sets the length of the keyboard click. Bigger n = longer click.
- `POKE 23692,255` (SCR CT) holds off the scroll? prompt for the next 255 scrolls; keep poking it with a number above 1 and the screen scrolls on without asking.
- `POKE 23658,8` turns CAPS LOCK on (FLAGS2 bit 3 = 1); `POKE 23658,0` turns it off.
- **FRAMES** 23672–23674 is a 3-byte frame counter, low byte first, +1 every 1/50 s. `PRINT (PEEK 23672+256*PEEK 23673+65536*PEEK 23674)/50` gives seconds since power-on (wraps after about 3 days 21 hours). Use 60 in 60 Hz countries.
- The ROM's IM 1 interrupt routine (MASK_INT at $0038) increments FRAMES and calls KEYBOARD, which updates KSTATE and LAST K. It bumps FRAMES' top byte with `INC (IY+$40)`, so it needs **IY = $5C3A** (ERR NR). Many ROM routines reach system variables through IY. A game running IM 2 that uses IY freely (like ours) stops these variables updating, and must restore IY = $5C3A and IM 1 before calling ROM/BASIC again.

Sources: ZX Spectrum BASIC manual, ch. 25 'The system variables' (worldofspectrum.org/ZXBasicManual/zxmanchap25.html) and ch. 18 · SkoolKid, The Complete Spectrum ROM Disassembly (skoolkid.github.io/rom/buffers/sysvars.html)